CLASS 10 · PART TEST-1 · STUDENT PERFORMANCE REPORT

Arnav Karthik Medichalam

Pragya G10 (EXT) Achievers · Adm No **246419493**

Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

227 / 296 — **76.7%**

16	12	4/65	4
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **188.9 / 300** (63.0%) → this test **76.7%** ▲ **13.7**

69 marks lost: 25 on 5 wrong answers (incl. 5 negative) · 44 on 11 skipped

JEE Advanced

09 Oct 2026 · 50 questions

109 / 176 — **61.9%**

94	65	15/61	22
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **98.2 / 180** (54.6%) → this test **61.9%** ▲ **7.4**

67 marks lost: 40 on 10 wrong answers (incl. 6 negative) · 24 on 7 skipped · 3 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 65 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 227 / 296 · 76.7%							
Maths	**63**/100	63%	**113**	58.2 / 100	**55.5**/100 55.5%	▲ **7.5**	16 / 1 / 8
Physics	**81**/100	81%	**38**	53.8 / 100	**74.6**/100 74.6%	▲ **6.4**	21 / 3 / 1
Chemistry	**83**/96	86%	**4**	55.9 / 91	**58.7**/100 58.7%	▲ **27.8**	21 / 1 / 2
JEE Advanced 109 / 176 · 61.9%							
Maths	**20**/60	33%	**503**	30.5 / 57	**31.5**/60 52.5%	▼ **19.2**	7 / 6 / 4
Physics	**36**/56	64%	**73**	24.5 / 56	**33.9**/60 56.5%	▲ **7.8**	11 / 4 / 1
Chemistry	**53**/60	88%	**12**	38.2 / 60	**32.8**/60 54.7%	▲ **33.6**	15 / 0 / 2

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Chemistry** is your strongest JEE Mains subject: 83/96 (86%), All-India subject rank 4.
2. **Maths** needs the most work in JEE Mains: 63/100 (63%), subject rank 113; you lost 9 of its 25 questions.
3. **Chemistry** is your strongest JEE Advanced subject: 53/60 (88%), All-India subject rank 12.
4. **Maths** needs the most work in JEE Advanced: 20/60 (33%), subject rank 503; you lost 10 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Advanced Maths** – 33% here against 52% on average.
6. Biggest improvement: **JEE Advanced Chemistry** – 88% here against 55% on average.
7. **1 question** you lost were answered correctly by 70%+ of the batch's top 100 – about **3 marks** within easy reach (Advanced Q3).
8. Negative marking cost you **5 in JEE Mains and 6 in JEE Advanced**.
9. You left **11 in JEE Mains and 7 in JEE Advanced** questions blank, and answered 5 in JEE Mains and 10 in JEE Advanced wrong – both are marks to recover.

JEE Mains – question by question

227 / 296 · 76.7%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	**5**	**6**	7	**8**	9	10	**11**	**12**	13	14	15	**16**	17	**18**	19	**20**	21	22	**23**	24	25
Physics	26	27	28	29	**30**	31	32	33	**34**	35	36	37	38	39	40	**41**	42	43	44	45	**46**	47	48	49	50
Chemistry	51	52	**53**	54	55	~~56~~	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	**73**	**74**	75

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 65 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

9 of 25 questions lost · **37** marks · ● 4 within reach · ● 5 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q23	Within reach	**Area Under Curves**	Numerical · +4 / −1	**Wrong**	5	**66%**	72%
Q8	Within reach	**Definite Integration**	Single correct · +4 / −1	**Skipped**	4	**56%**	38%
Q12	Within reach	**Indefinite Integration**	Single correct · +4 / −1	**Skipped**	4	**51%**	43%
Q5	Within reach	**Definite Integration**	Single correct · +4 / −1	**Skipped**	4	**40%**	32%
Q18	Tough one	**Definite Integration** – ∫ of min{|x − 3k|} – periodic zig-zag	Single correct · +4 / −1	**Skipped**	4	**37%**	40%
Q6	Tough one	**Indefinite Integration** – ∫ $\sqrt{(x^2+1)}$ [ln(x^2+1) − 2 ln x] / x^4 dx	Single correct · +4 / −1	**Skipped**	4	**33%**	31%
Q11	Tough one	**Definite Integration**	Single correct · +4 / −1	**Skipped**	4	**29%**	26%
Q20	Tough one	**Definite Integration** – King's property with f(x) + f(y) = f(xy)	Single correct · +4 / −1	**Skipped**	4	**29%**	32%
Q16	Tough one	**Definite Integration** – $\int_0^{100}$ ([$\tan^{-1}$x] + [$\cot^{-1}$x]) dx with GIF	Single correct · +4 / −1	**Skipped**	4	**16%**	11%

Physics

4 of 25 questions lost · **19** marks · ● 2 within reach · ● 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q41	Within reach	**Kinematics**	Single correct · +4 / −1	**Wrong**	5	**65%**	71%
Q34	Within reach	**Current Electricity**	Single correct · +4 / −1	**Wrong**	5	**45%**	37%
Q46	Tough one	**Electrostatics**	Numerical · +4 / −1	**Wrong**	5	**25%**	25%
Q30	Tough one	**Electrostatics** – Field energy inside vs outside a charged sphere	Single correct · +4 / −1	**Skipped**	4	**17%**	11%

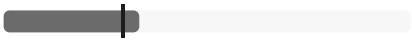
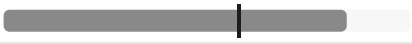
Chemistry

3 of 24 questions lost · **13** marks · ● 2 within reach · ● 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q53	Within reach	**Alcohols, Phenols & Ethers**	Single correct · +4 / −1	**Skipped**	4	**68%**	69%
Q73	Within reach	**Alcohols, Phenols & Ethers**	Numerical · +4 / −1	**Skipped**	4	**43%**	49%
Q74	Tough one	**Alcohols, Phenols & Ethers** – Site of benzylation of 2-naphthoxide	Numerical · +4 / −1	**Wrong**	5	**12%**	15%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Definite Integration	9	**12**/36		29%	**Q5**, **Q8**, **Q11**, **Q16**, **Q18**, **Q20**
Indefinite Integration	7	**20**/28		61%	**Q6**, **Q12**
Area Under Curves	5	**15**/20		86%	**Q23**
Differential Equations	4	**16**/16		84%	– none
Physics					
Electrostatics	8	**23**/32		27%	**Q30**, **Q46**
Current Electricity	8	**27**/32		57%	**Q34**
Kinematics	9	**31**/36		75%	**Q41**
Chemistry					
Alcohols, Phenols & Ethers	10	**27**/40		48%	**Q53**, **Q73**, **Q74**
Aldehydes & Ketones	14	**56**/56		65%	– none

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

9 questions · easy misses first
cropped from the question paper

Q23 Area Under Curves

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **72%** (47 of 65)

Within reach
You: Wrong

23. The area bounded by the lines $y = ||x-1|-2|$ with x-axis is _____ (in sq.units)

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8 Definite Integration

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **56%** got it right · Campus: **38%** (25 of 65)

Within reach
You: Skipped

8. If $\int_0^{\frac{\pi}{2}} \frac{\tan x}{\tan x + \sec x} dx = m(\pi + n)$, then $m \times n$ is equal to

A) $-\frac{1}{2}$ B) 1 C) -1 D) $\frac{1}{2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12 Indefinite Integration

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **51%** got it right · Campus: **43%** (28 of 65)

Within reach
You: Skipped

12. $\int \sqrt{\sec x - 1}\, dx =$ _____(c is arbitrary constant)

A) $2\ln\left|\cos\frac{x}{2} + \sqrt{\cos^2\frac{x}{2} + \frac{1}{2}}\right| + c$ B) $\ln\left|\cos\frac{x}{2} - \sqrt{\cos^2\frac{x}{2} - \frac{1}{2}}\right| + c$

C) $-2\ln\left|\cos\frac{x}{2} + \sqrt{\cos^2\frac{x}{2} - \frac{1}{2}}\right| + c$ D) $\ln\left|\cos\frac{x}{2} - \sqrt{\cos^2\frac{x}{2} + \frac{1}{2}}\right| + c$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q5

Definite Integration
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **40%** got it right · Campus: **32%** (21 of 65)

Within reach
You: Skipped

5. $\int_{-1}^{1} \sin x \ln\left(1^x + 2^x + 3^x + 6^x\right) dx =$

A) $(\ln 6)\left(\cos\left(\frac{\pi+4}{4}\right)\right)$

B) $(\ln 6)\left(\sin\left(\frac{\pi+4}{4}\right)\right)$

C) $\sqrt{2}(\ln 6)\left(\sin\left(\frac{4-\pi}{4}\right)\right)$

D) $\sqrt{2}(\ln 6)\left(\sin\left(\frac{\pi+4}{4}\right)\right)$

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18

Definite Integration – ∫ of min{|x − 3k|} – periodic zig-zag
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **37%** got it right · Campus: **40%** (26 of 65)

Tough one
You: Skipped

18. Let $f(x) = \min\{\ldots, |x+6|, |x+3|, |x|, |x-3|, |x-6|, \ldots\}$ then $\int_{-4}^{3} f(x)\,dx =$ _____

A) 4

B) $\frac{9}{2}$

C) 5

D) $\frac{11}{2}$

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q6

Indefinite Integration – ∫ √(x²+1)[ln(x²+1) − 2 ln x] / x⁴ dx
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **33%** got it right · Campus: **31%** (20 of 65)

Tough one
You: Skipped

6. $\int \frac{\sqrt{x^2+1}\left[\ln\left(x^2+1\right) - 2\ln x\right]}{x^4} dx$ is equal to (c is arbitrary constant)

A) $\frac{\left(x^2+1\right)^{3/2}}{x^3}\left[\frac{2}{3} - \ln\left(\frac{x^2+1}{x^2}\right)\right] + c$

B) $\frac{\left(x^2+1\right)^{3/2}}{3x^3}\left[\ln\left(\frac{x^2+1}{x^2}\right) - \frac{2}{3}\right] + c$

C) $\frac{2\left(x^2+1\right)^{3/2}}{3x^3}\left[\frac{1}{3} - \frac{1}{2}\ln\left(\frac{x^2+1}{x^2}\right)\right] + c$

D) $\frac{\left(x^2+1\right)^{3/2}}{3x^3}\left[\ln\left(\frac{x^2+1}{x^2}\right) + \frac{2}{3}\right] + c$

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q11

Definite Integration
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **29%** got it right · Campus: **26%** (17 of 65)

Tough one
You: Skipped

11. Let $f(x)=\sin x+\sum_{r=0}^{2}\cos\left(x+\frac{2r\pi}{3}\right)$, $g(x)=\cos x+\sum_{r=0}^{2}\sin\left(x+\frac{2r\pi}{3}\right)$ then $\int_{0}^{\frac{\pi}{2}} f(g(x))f(x)\,dx=$

A) $1+\sin 1$ B) $1-\sin 1$ C) $1+\cos 1$ D) $1-\cos 1$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Definite Integration – King's property with f(x) + f(y) = f(xy)
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **29%** got it right · Campus: **32%** (21 of 65)

Tough one
You: Skipped

20. Let $f:(0,\infty)\to R$ be a continuous function, satisfying $f(x)+f(y)=f(xy)$. Then the value of $\int_{\sqrt{f(e)}}^{\sqrt{f(e^2)}}\frac{x\tan x^2}{\tan x^2+\tan\left(f(e^3)-x^2\right)}dx$ is equal to

A) $\frac{1}{2}f(e^2)$ B) $\frac{1}{4}f(e^2)$ C) $\frac{1}{2}f(e)$ D) $\frac{f(e)}{4}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16

Definite Integration – $\int_0^{100}([\tan^{-1}x]+[\cot^{-1}x])\,dx$ with GIF
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **16%** got it right · Campus: **11%** (7 of 65)

Tough one
You: Skipped

16. If $\int_{0}^{100}\left(\left[\tan^{-1}x\right]+\left[\cot^{-1}x\right]\right)dx=\ldots$ [.] denotes the greatest integer function

A) $100-\cot 1$ B) $100+2\cot 1$ C) $100-2\cot 2$ D) $100+2\cot 2$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

4 questions · easy misses first
cropped from the question paper

Q41 Kinematics

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **65%** got it right · Campus: **71%** (46 of 65)

Within reach
You: Wrong

41. A swimmer can swim at 5 m/s in still water. A river 120 m wide flows at 3 m/s. If the swimmer heads so as to reach the point directly opposite on the other bank, the time taken is

A) 24 s B) 40 s C) 15 s D) 30 s

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34 Current Electricity

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **45%** got it right · Campus: **37%** (24 of 65)

Within reach
You: Wrong

34. In a Wheatstone bridge ABCD, a 12 V battery of negligible internal resistance is connected between A and C. The arms are AB = 2 Ω, BC = 3 Ω, AD = 3 Ω and DC = 3 Ω. An ammeter of resistance 0.3 Ω is connected between B and D. The current through the ammeter is

A) 0.2 A B) 0.6 A C) 0.4 A D) 1.2 A

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q46 Electrostatics

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5
Top 100: **25%** got it right · Campus: **25%** (16 of 65)

Tough one
You: Wrong

46. In a region the electric field is $E = \left(5x\hat{i} + 3y\hat{j} + 4\hat{k}\right) N/C$, with x, y in meters. A cube of side 1m has one vertex at the origin and its edges along the positive x, y and z axis. The net electric flux through the cube (in Nm^2/C) is _____.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30 **Electrostatics – Field energy inside vs outside a charged sphere**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **17%** got it right · Campus: **11%** (7 of 65)

Tough one
You: Skipped

30. A solid sphere of radius R carries a total charge Q distributed uniformly throughout its volume. The ratio of the electrostatic field energy stored inside the sphere to that stored outside the sphere is

A) 3 : 5 B) 1 : 5 C) 1 : 3 D) 1 : 2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry — questions to revisit

3 questions · easy misses first
cropped from the question paper

Q53 Alcohols, Phenols & Ethers

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **68%** got it right · Campus: **69%** (45 of 65)

Within reach
You: Skipped

53. Phenol industrially prepared from

A) Benzene sulphonic acid
B) Cumene
C) Benzene diazonium chloride
D) Chlorobenzene

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q73 Alcohols, Phenols & Ethers

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **43%** got it right · Campus: **49%** (32 of 65)

Within reach
You: Skipped

73. How many chiral alcohols, including stereoisomers can give 2-chloro-2-methylbutane as the major product on reaction with $NaCl / H_2SO_4$ by hydride ion shift during its path?

Why did I lose this question? (write the reason and the correct approach)

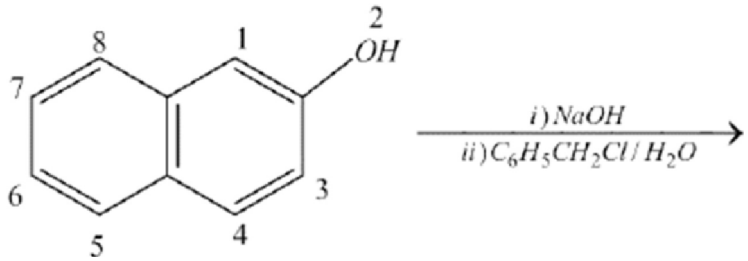
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74 Alcohols, Phenols & Ethers — Site of benzylation of 2-naphthoxide

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5
Top 100: **12%** got it right · Campus: **15%** (10 of 65)

Tough one
You: Wrong

74. Which site is predominately benzylated in the given reaction?



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced – question by question

109 / 176 · 61.9%
09 Oct 2026

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Physics	18	19	20	21	22	23	24	~~25~~	26	27	28	29	30	31	32	33	34
Chemistry	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 61 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

10 of 17 questions lost · **40** marks · ● 1 easy miss · ● 8 within reach · ● 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q3	Easy miss	**Area Under Curves**	Single correct · +3 / −1	Skipped	3	85%	87%
Q6	Within reach	**Area Under Curves** – Line x = α halving the area between x^3 and x	Multiple correct · +4 / −2	Wrong	6	68%	69%
Q1	Within reach	**Indefinite Integration**	Single correct · +3 / −1	Wrong	4	66%	70%
Q13	Within reach	**Definite Integration**	Integer · +4 / 0	Skipped	4	64%	70%
Q16	Within reach	**Area Under Curves**	Paragraph · +3 / 0	Wrong	3	48%	41%
Q7	Within reach	**Differential Equations** – Curve from tangent-intercept ratio BP : AP = 3 : 1	Multiple correct · +4 / −2	Wrong	6	43%	26%
Q2	Within reach	**Definite Integration** – $f(a) = \int \ln(1+x^a)/(1+x)\,dx$ – find f(4) + f(1/4)	Single correct · +3 / −1	Wrong	4	42%	34%
Q17	Within reach	**Differential Equations**	Paragraph · +3 / 0	Skipped	3	42%	33%
Q15	Within reach	**Definite Integration**	Paragraph · +3 / 0	Wrong	3	41%	39%
Q11	Tough one	**Differential Equations**	Integer · +4 / 0	Skipped	4	25%	21%

Physics

5 of 16 questions lost · **17** marks (+3 on partly-correct answers) · ● 2 within reach · ● 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q29	Within reach	**Electrostatics**	Integer · +4 / 0	Wrong	4	54%	34%
Q33	Within reach	**Electrostatics** – Flux through a ring – solid angle	Paragraph · +3 / 0	Wrong	3	42%	38%
Q34	Tough one	**Electrostatics** – Flux through a cube from a wire along one edge	Paragraph · +3 / 0	Wrong	3	33%	23%
Q20	Tough one	**Kinematics** – Shell fired from a plateau – minimum landing distance	Single correct · +3 / −1	Skipped	3	29%	23%
Q27	Tough one	**Kinematics** – Projection angle that maximises area under the trajectory	Integer · +4 / 0	Wrong	4	27%	23%

Chemistry

2 of 17 questions lost · 7 marks · ● 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q50	Tough one	**Carboxylic Acids** – Molar mass of an acid from NaOH titration	Paragraph · +3 / 0	**Skipped**	3	**36%**	52%
Q46	Tough one	**Alcohols, Phenols & Ethers** – Identifying polar aprotic solvents	Integer · +4 / 0	**Skipped**	4	**24%**	8%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Area Under Curves	3	≈ **-2**/10		60%	**Q3**, **Q6**, **Q16**
Differential Equations	5	≈ **5**/18		30%	**Q7**, **Q11**, **Q17**
Definite Integration	5	≈ **6**/17		54%	**Q2**, **Q13**, **Q15**
Indefinite Integration	4	≈ **11**/15		71%	**Q1**
Physics					
Electrostatics	6	≈ **11**/21		45%	**Q29**, **Q33**, **Q34**
Kinematics	8	≈ **21**/28		41%	**Q20**, **Q27**
Current Electricity	2	≈ **7**/7		68%	– none
Chemistry					
Carboxylic Acids	4	≈ **11**/14		48%	**Q50**
Alcohols, Phenols & Ethers	7	≈ **22**/26		62%	**Q46**
Aldehydes & Ketones	6	≈ **20**/20		83%	– none

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

10 questions · easy misses first
cropped from the question paper

Q3 — Area Under Curves

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 3
Top 100: **85%** got it right · Campus: **87%** (53 of 61)

Easy miss
You: Skipped

3. The area bounded by the curve $4y^2 = x^2\left(4 - x^2\right)$ is

A) 7/3 sq.units B) 10/3 sq.units C) 13/3 sq.units D) 16/3 sq.units

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q6 — Area Under Curves – Line x = α halving the area between x³ and x

JEE Advanced · Maths · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **68%** got it right · Campus: **69%** (42 of 61)

Within reach
You: Wrong

6. If the line x=α divides the area of the region $R = \left\{(x, y) \in R^2 : x^3 \le y \le x, 0 \le x \le 1\right\}$ into two equal parts, then

A) $\frac{1}{2} \le \alpha \le 1$ B) $\alpha^4 + 4\alpha^2 - 1 = 0$ C) $0 \le \alpha \le \frac{1}{2}$ D) $2\alpha^4 - 4\alpha^2 + 1 = 0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q1

Indefinite Integration

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4

Top 100: **66%** got it right · Campus: **70%** (43 of 61)

Within reach

You: Wrong

1. $\int \frac{2e^{2x}+e^{x}}{3e^{2x}-2e^{x}-1}dx$ is equal to

A) $\frac{1}{4}\ln\left|\frac{\left(e^{x}-1\right)^{1/3}}{\left(3e^{x}+1\right)^{3}}\right|+C$

B) $\frac{1}{4}\ln\left|\frac{\left(e^{x}-1\right)^{3}}{\left(3e^{x}+1\right)^{1/3}}\right|+C$

C) $\frac{1}{3}\ln\left|\frac{\left(e^{x}-1\right)^{4}}{\left(3e^{x}+1\right)^{1/4}}\right|+C$

D) $\frac{1}{3}\ln\left|\frac{\left(e^{x}-1\right)^{1/4}}{\left(3e^{x}+1\right)^{4}}\right|+C$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13

Definite Integration

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **64%** got it right · Campus: **70%** (43 of 61)

Within reach

You: Skipped

13. Let $t(x)=\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)(x+4)}}}$ then $\int_0^{100} t(x)dx$ is K. Find $\frac{K}{1700}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16

Area Under Curves

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **48%** got it right · Campus: **41%** (25 of 61)

Within reach

You: Wrong

PARAGRAPH-II

Let $f(x)$ be a differentiable function.

$f(x)>0\forall x\in R \quad f(x)=\left(2\int_0^x f(t)dt\right)+1\forall x\in R$

16. Find the area bounded by $y=\ln(f(x))$, *when* $x=f(0)$ *and* $x=0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q7

Differential Equations – Curve from tangent-intercept ratio BP : AP = 3 : 1

JEE Advanced · Maths · Multiple correct · marking +4 / −2 · marks lost 6

Top 100: **43%** got it right · Campus: **26%** (16 of 61)

Within reach

You: Wrong

7. A curve $y = f(x)$ passes through the point (1,1) and tangent at P(x,y) cuts the x-axis and y-axis at A and B, respectively such that BP:AP=3:1, then

A) Equation of the curve is $xy^3 = 1$

B) Equation of normal at the point (1,1) is $x - 3y + 2 = 0$

C) Curve passes through the point $\left(\frac{1}{8}, 2\right)$

D) Equation of the curve is $x\frac{dy}{dx} + 3y = 0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q2

Definite Integration – f(a) = ∫ ln(1+x^a)/(1+x) dx – find f(4) + f(1/4)

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4

Top 100: **42%** got it right · Campus: **34%** (21 of 61)

Within reach

You: Wrong

2. Let $f(a) = \int_0^1 \frac{\ln(1+x^a)}{1+x} dx$ then $f(4) + f\left(\frac{1}{4}\right)$ is equal to

A) $\ln^2 1$ B) $\ln^2 2$ C) $\ln^2 3$ D) $\ln^2 4$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17 Differential Equations

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **42%** got it right · Campus: **33%** (20 of 61)

Within reach

You: Skipped

PARAGRAPH-II

Let $f(x)$ be a differentiable function.

$$f(x)>0\forall x\in R \qquad f(x)=\left(2\int_0^x f(t)dt\right)+1\forall x\in R$$

17. Find the value of $\lim_{x\to 0}\dfrac{f(x)-2x-1}{1-\cos x}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15 Definite Integration

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **41%** got it right · Campus: **39%** (24 of 61)

Within reach

You: Wrong

PARAGRAPH-I

$$\int_0^a f(x)dx=\int_0^a f(a-x)dx;\quad \int_a^b f(x)dx=\int_a^b f(a+b-x)dx;\quad \int_0^{2a} f(x)dx=\int_0^a\left(f(x)+f(2a-x)\right)dx$$

15. If $\left|\int_0^{\frac{\pi}{\pi}}\ln(1+\cos x)\,dx\right|=\dfrac{\pi}{q}\ln(p),(p,q\in N)$ then find the least possible value p+q

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q11 Differential Equations

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **25%** got it right · Campus: **21%** (13 of 61)

Tough one

You: Skipped

11. Let $e^{-x}\dfrac{d^2y}{dx^2}-\dfrac{dy}{dx}-y=0$ such that at the point $(0,1)$, $\dfrac{dy}{dx}=1$. Then the number of point of intersection of the given curve with $y=e^{e^x}$ is

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29 **Electrostatics**

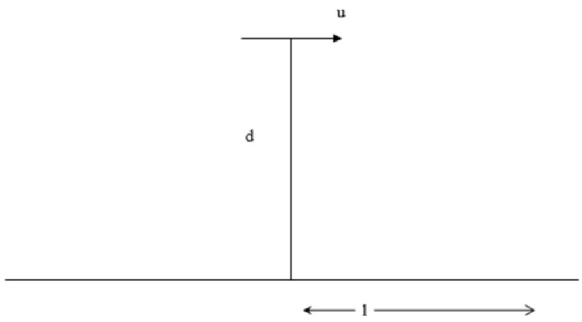
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **54%** got it right · Campus: **34%** (21 of 61)

Within reach

You: Wrong

29. An electron is projected from a distance d and with initial velocity u parallel to a uniformly charge flat conducting plate as shown. It strikes the plate after traveling a distance l along the direction of projection. The surface charge density of the conducting plate is $\sigma = \frac{nd\varepsilon_0 mu^2}{el^2}$, The value of n is:



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33 **Electrostatics – Flux through a ring – solid angle**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **42%** got it right · Campus: **38%** (23 of 61)

Within reach
You: Wrong

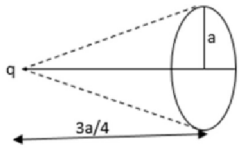
PARAGRAPH-II

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

33.



The flux associated with the ring due to the charge q is given by Xq/ε_0. The value of X is approx.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34 **Electrostatics – Flux through a cube from a wire along one edge**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **33%** got it right · Campus: **23%** (14 of 61)

Tough one
You: Wrong

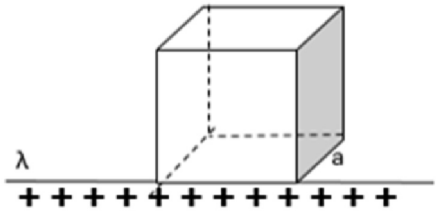
PARAGRAPH-II

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

34.



A cube of edge length a contains an infinite wire of linear charge density λ at one of edges the flux associated with cube is given by $P\lambda a/\varepsilon_0$.The value of P upto two decimal places is :

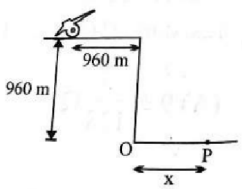
Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20 **Kinematics – Shell fired from a plateau – minimum landing distance**
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 3
Top 100: **29%** got it right · Campus: **23%** (14 of 61)

Tough one
You: Skipped

20. A gun is mounted on a plateau 960m away from its edge as shown. Height of plateau is 960m. The gun can fire shells with a velocity 100 m/s at any angle of the following choices. What is the minimum distance OP(=x) from the edge of plateau where shell of



gun can reach? (given $g=10m/s^2$, $\sin 74^0 = \sin 106^0 \simeq 0.96$)

A) 480 m B) 720 m C) 360 m D) 960 m

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Kinematics – Projection angle that maximises area under the trajectory

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **27%** got it right · Campus: **23%** (14 of 61)

Tough one

You: Wrong

27. A projectile is thrown with a speed 10 m/s, at an angle θ with the horizontal. For a certain value of angle of projection $\theta = \frac{(n+1)\pi}{12}$ the area bound by the trajectory with horizontal is maximum value of n is (g=10 m/s^2). Assume no airdrag.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q50 **Carboxylic Acids** — **Molar mass of an acid from NaOH titration**
JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **36%** got it right · Campus: **52%** (32 of 61)

Tough one
You: Skipped

PARAGRAPH-II

Carboxylic acids are organic compounds containing the carbonyl(>C=O) and hydroxyl (-OH) functional groups combined as a carboxyl group(-COOH). Due to resonance stabilization of the carboxylate anion, carboxylic acids exhibit higher acidity compared to alcohols and phenols. They participate in several characteristic reactions, including nucleophilic acyl substitution (esterification, amide formation, acid halide synthesis), decarboxylation, and reduction to primary alcohols. Furthermore, the presence of the –COOH group strongly influences physical properties, enabling extensive intermolecular hydrogen bonding that results in higher boiling points and stable dimeric structures in non-polar solvents.

50. A 0.600 g sample of a pure monoprotic carboxylic acid (R-COOH) requires 25.00 mL of 0.200 M NaOH for complete neutralization. Calculate the molar mass of the carboxylic acid?

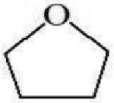
Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

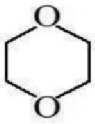
Q46 **Alcohols, Phenols & Ethers** – **Identifying polar aprotic solvents**
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4
Top 100: **24%** got it right · Campus: **8%** (5 of 61)

Tough one
You: Skipped

46. How many of the following are polar aprotic solvents?



Diethyl ether, Benzene, Ethyl alcohol,



Petrol

[P] [Q] [R] [S] [T] [U]

Acetone DMSO (dimethyl sulphoxide DMF (dimethyl formamide formic acid

[Q] [W] [X] [Y]

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q56 — JEE Advanced: Q25. **Program rank** = rank in Pragya G10 (EXT) Achievers across all campuses, from the batch's question-wise sheet. **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 3 marks went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.